

✓ Exploratory Data Analysis (EDA) on the Titanic Dataset

Objective: Analyze the Titanic dataset using Python to identify trends, patterns, relationships, and insights through statistical analysis and data visualization.

Tools Used:

- Python
- Pandas
- Matplotlib
- Seaborn

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

# Display plots inside the notebook
%matplotlib inline

# Set plot style
sns.set_style("whitegrid")

# Load the Titanic dataset
df = pd.read_csv("train.csv")

# Display the first 5 rows
df.head()
```

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked
0	1	0	3	Braund, Mr. Owen Harris	male	22.0	1	0	A/5 21171	7.2500	NaN	S
1	2	1	1	Cumings, Mrs. John Bradley (Florence Briggs Th...)	female	38.0	1	0	PC 17599	71.2833	C85	S
2	3	1	3	Heikkinen, Miss. L. S.	female	26.0	0	0	STON/O2. 3101282	7.9250	NaN	S

```
# Dataset dimensions
print("Dataset Shape:", df.shape)

# Column names
print("\nColumns:")
print(df.columns)

# Dataset information
print("\nDataset Information:")
df.info()
```

```

Dataset Shape: (891, 12)

Columns:
Index(['PassengerId', 'Survived', 'Pclass', 'Name', 'Sex', 'Age', 'SibSp',
      'Parch', 'Ticket', 'Fare', 'Cabin', 'Embarked'],
      dtype='object')

Dataset Information:
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 891 entries, 0 to 890
Data columns (total 12 columns):
#   Column          Non-Null Count  Dtype
---  -
0   PassengerId      891 non-null    int64
1   Survived         891 non-null    int64
2   Pclass           891 non-null    int64
3   Name             891 non-null    object
4   Sex              891 non-null    object
5   Age              714 non-null    float64
6   SibSp            891 non-null    int64
7   Parch            891 non-null    int64
8   Ticket           891 non-null    object
9   Fare             891 non-null    float64
10  Cabin            204 non-null    object
11  Embarked         889 non-null    object
dtypes: float64(2), int64(5), object(5)
memory usage: 83.7+ KB

```

Observation

- The Titanic dataset contains passenger details such as age, gender, ticket class, fare, and survival status.
- Some columns contain missing values that will be explored later.
- The dataset includes both numerical and categorical features for analysis.

```

# Statistical summary of numerical columns
df.describe()

```

	PassengerId	Survived	Pclass	Age	SibSp	Parch	Fare
count	891.000000	891.000000	891.000000	714.000000	891.000000	891.000000	891.000000
mean	446.000000	0.383838	2.308642	29.699118	0.523008	0.381594	32.204208
std	257.353842	0.486592	0.836071	14.526497	1.102743	0.806057	49.693429
min	1.000000	0.000000	1.000000	0.420000	0.000000	0.000000	0.000000
25%	223.500000	0.000000	2.000000	20.125000	0.000000	0.000000	7.910400
50%	446.000000	0.000000	3.000000	28.000000	0.000000	0.000000	14.454200
75%	668.500000	1.000000	3.000000	38.000000	1.000000	0.000000	31.000000
max	891.000000	1.000000	3.000000	80.000000	8.000000	6.000000	512.329200

Observation

- The statistical summary provides an overview of numerical features such as Age and Fare.
- There is noticeable variation in passenger ages and ticket fares.

```
# Check missing values
df.isnull().sum()
```

	0
PassengerId	0
Survived	0
Pclass	0
Name	0
Sex	0
Age	177
SibSp	0
Parch	0
Ticket	0
Fare	0
Cabin	687
Embarked	2

dtype: int64

Observation

- Missing values are present in Age, Cabin, and Embarked
- Cabin has the highest number of missing values.

```
print("Passenger Class")
print(df["Pclass"].value_counts())

print("\nGender")
print(df["Sex"].value_counts())

print("\nSurvival")
print(df["Survived"].value_counts())
```

```
Passenger Class
Pclass
3    491
1    216
2    184
Name: count, dtype: int64
```

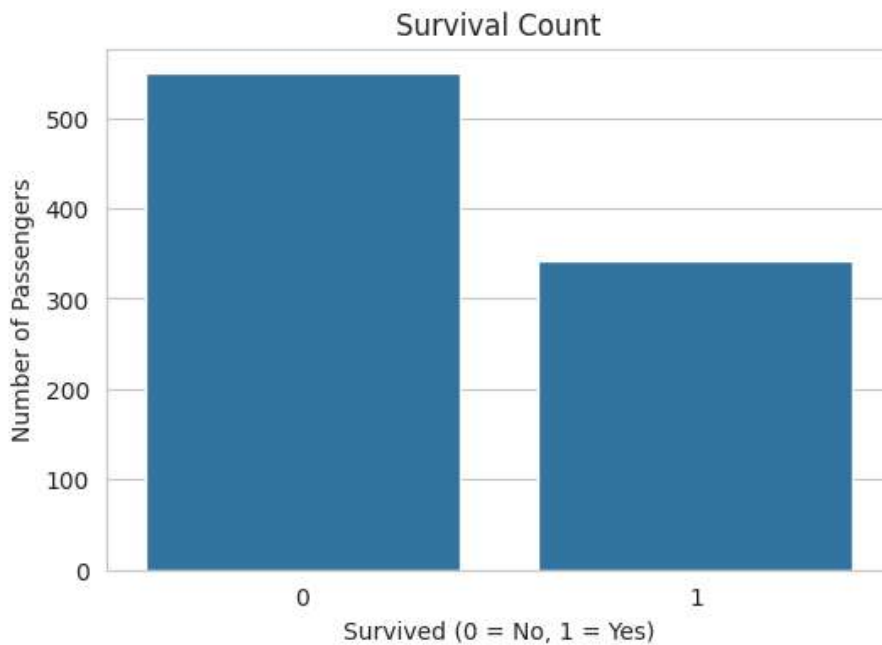
```
Gender
Sex
male    577
female  314
Name: count, dtype: int64
```

```
Survival
Survived
0    549
1    342
Name: count, dtype: int64
```

Observation

- Most passengers traveled in Third Class.
- Male passengers outnumber female passengers.
- More passengers did not survive than survived.

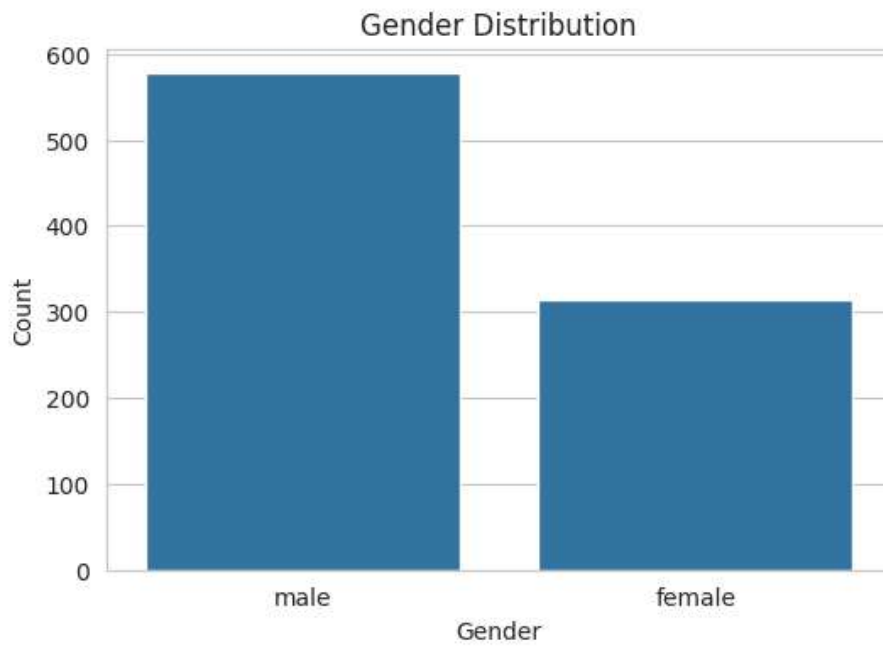
```
plt.figure(figsize=(6,4))
sns.countplot(x='Survived', data=df)
plt.title("Survival Count")
plt.xlabel("Survived (0 = No, 1 = Yes)")
plt.ylabel("Number of Passengers")
plt.show()
```



Observation

- More passengers did not survive than survived, indicating the disaster had a high mortality rate.

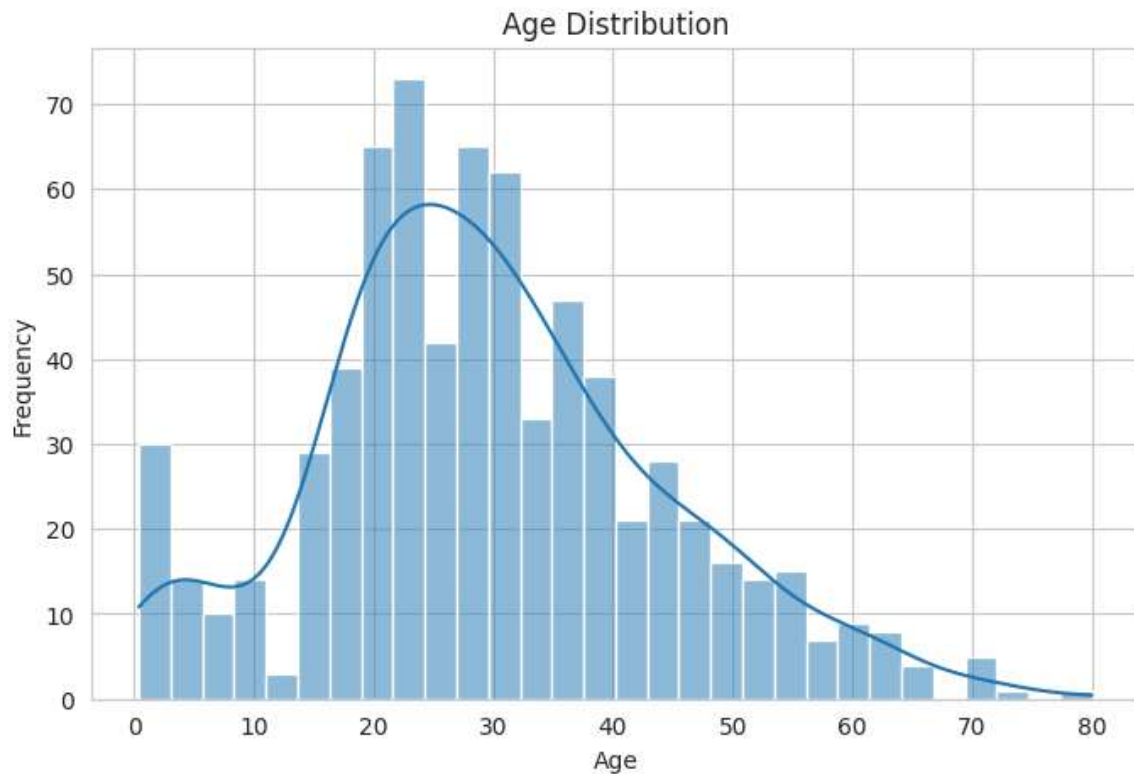
```
plt.figure(figsize=(6,4))
sns.countplot(x='Sex', data=df)
plt.title("Gender Distribution")
plt.xlabel("Gender")
plt.ylabel("Count")
plt.show()
```



Observation

- The dataset contains more male passengers than female passengers.

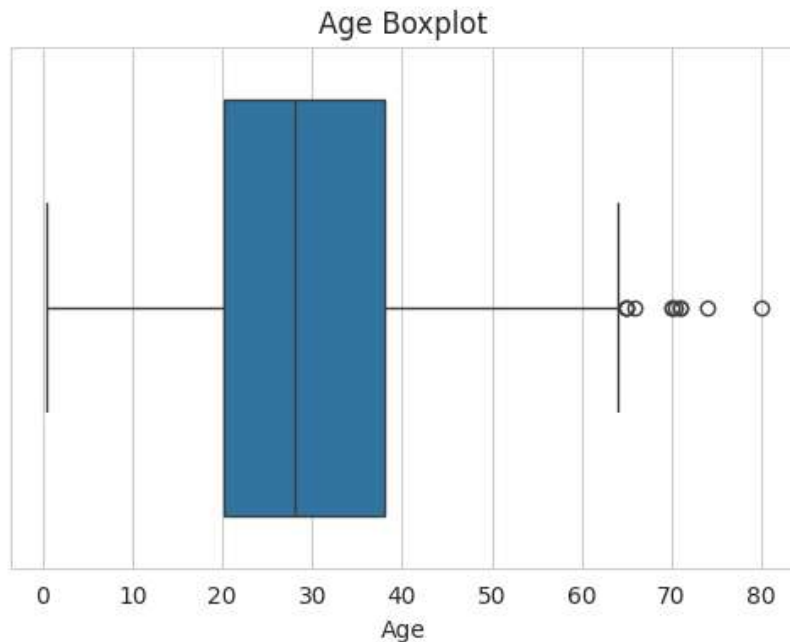
```
plt.figure(figsize=(8,5))
sns.histplot(df['Age'], bins=30, kde=True)
plt.title("Age Distribution")
plt.xlabel("Age")
plt.ylabel("Frequency")
plt.show()
```



Observation

- Most passengers were between 20 and 40 years of age, with fewer elderly passengers.

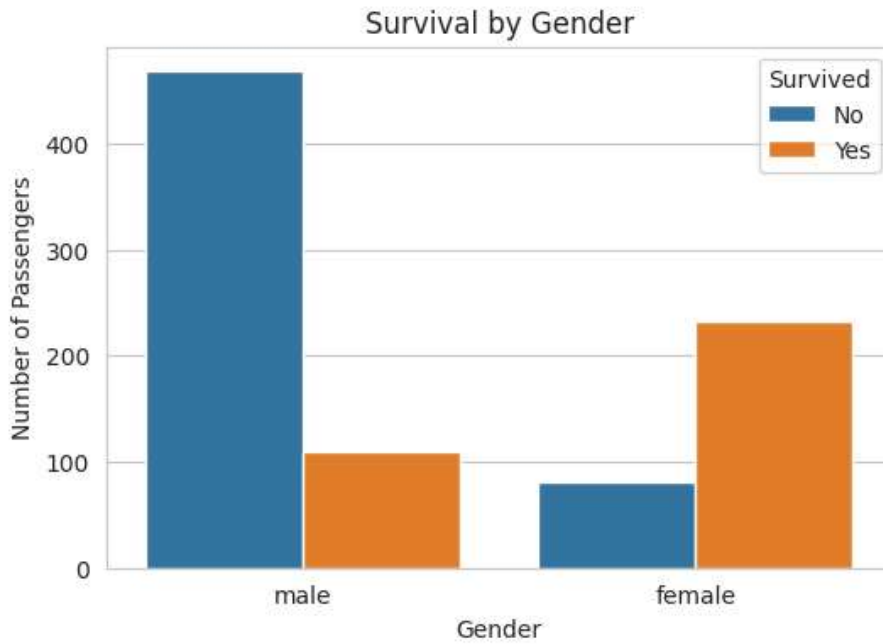
```
plt.figure(figsize=(6,4))
sns.boxplot(x=df['Age'])
plt.title("Age Boxplot")
plt.show()
```



Observation

- The age distribution contains several outliers, representing unusually old or young passengers.

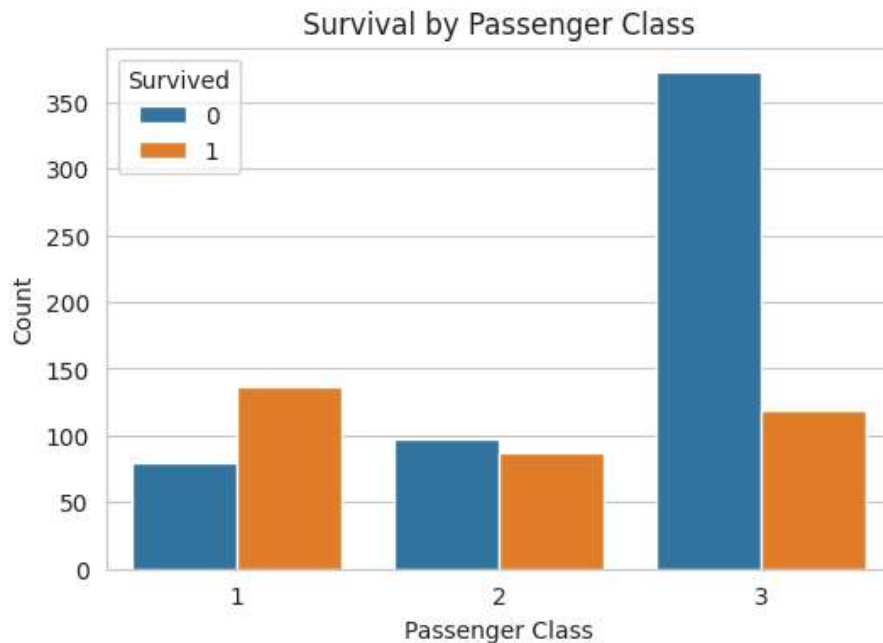
```
plt.figure(figsize=(6,4))
sns.countplot(x='Sex', hue='Survived', data=df)
plt.title("Survival by Gender")
plt.xlabel("Gender")
plt.ylabel("Number of Passengers")
plt.legend(title="Survived", labels=["No", "Yes"])
plt.show()
```



Observation

- Female passengers had a significantly higher survival rate than male passengers.

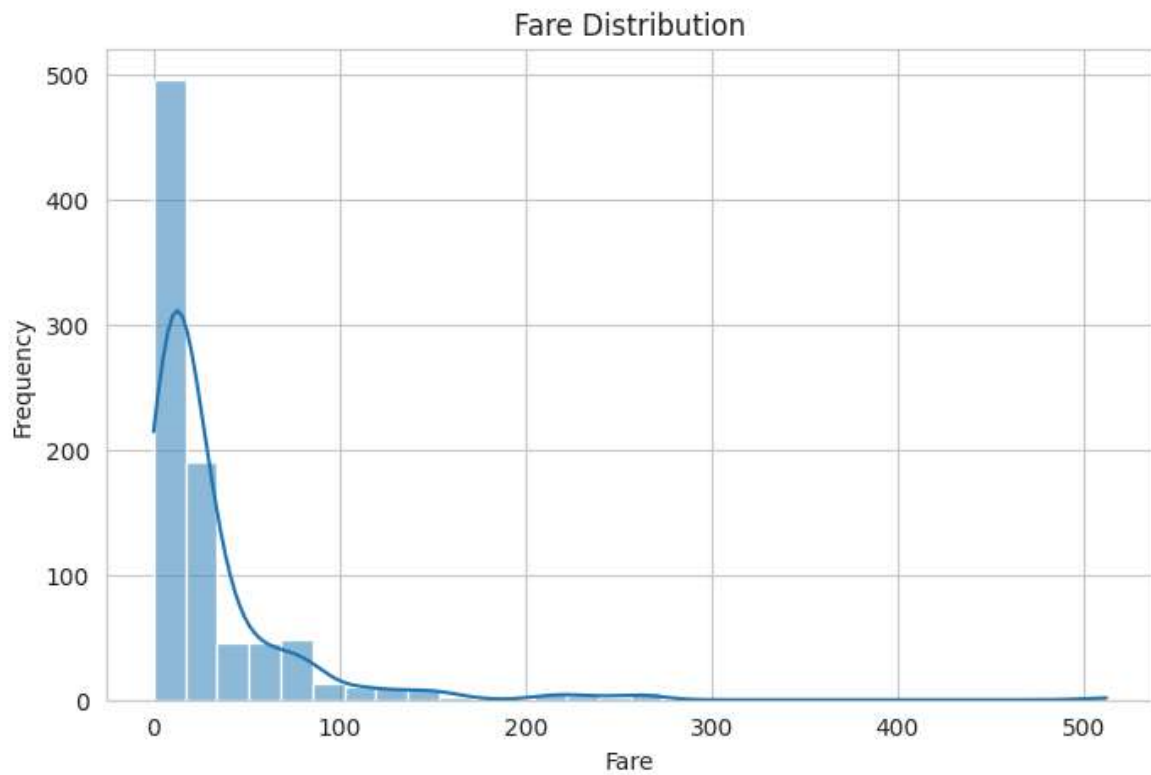
```
plt.figure(figsize=(6,4))
sns.countplot(x='Pclass', hue='Survived', data=df)
plt.title("Survival by Passenger Class")
plt.xlabel("Passenger Class")
plt.ylabel("Count")
plt.show()
```



Observation

- First-class passengers had the highest survival rate, while third-class passengers had the lowest.

```
plt.figure(figsize=(8,5))
sns.histplot(df['Fare'], bins=30, kde=True)
plt.title("Fare Distribution")
plt.xlabel("Fare")
plt.ylabel("Frequency")
plt.show()
```



Observation

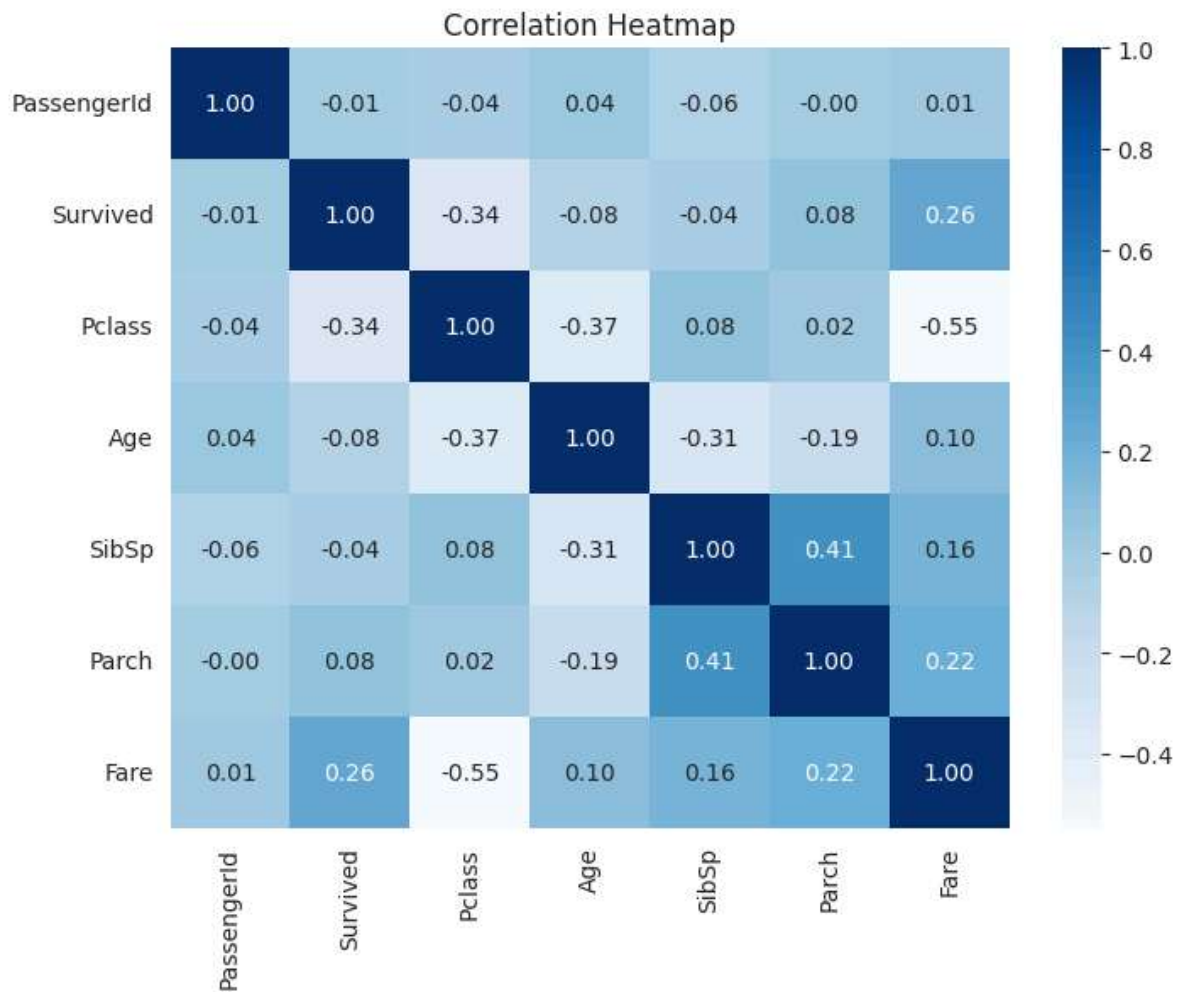
- Most passengers paid lower fares, while only a few passengers paid very high ticket prices.

```
plt.figure(figsize=(8,6))

numeric_df = df.select_dtypes(include=['number'])

sns.heatmap(numeric_df.corr(), annot=True, cmap='Blues', fmt=".2f")

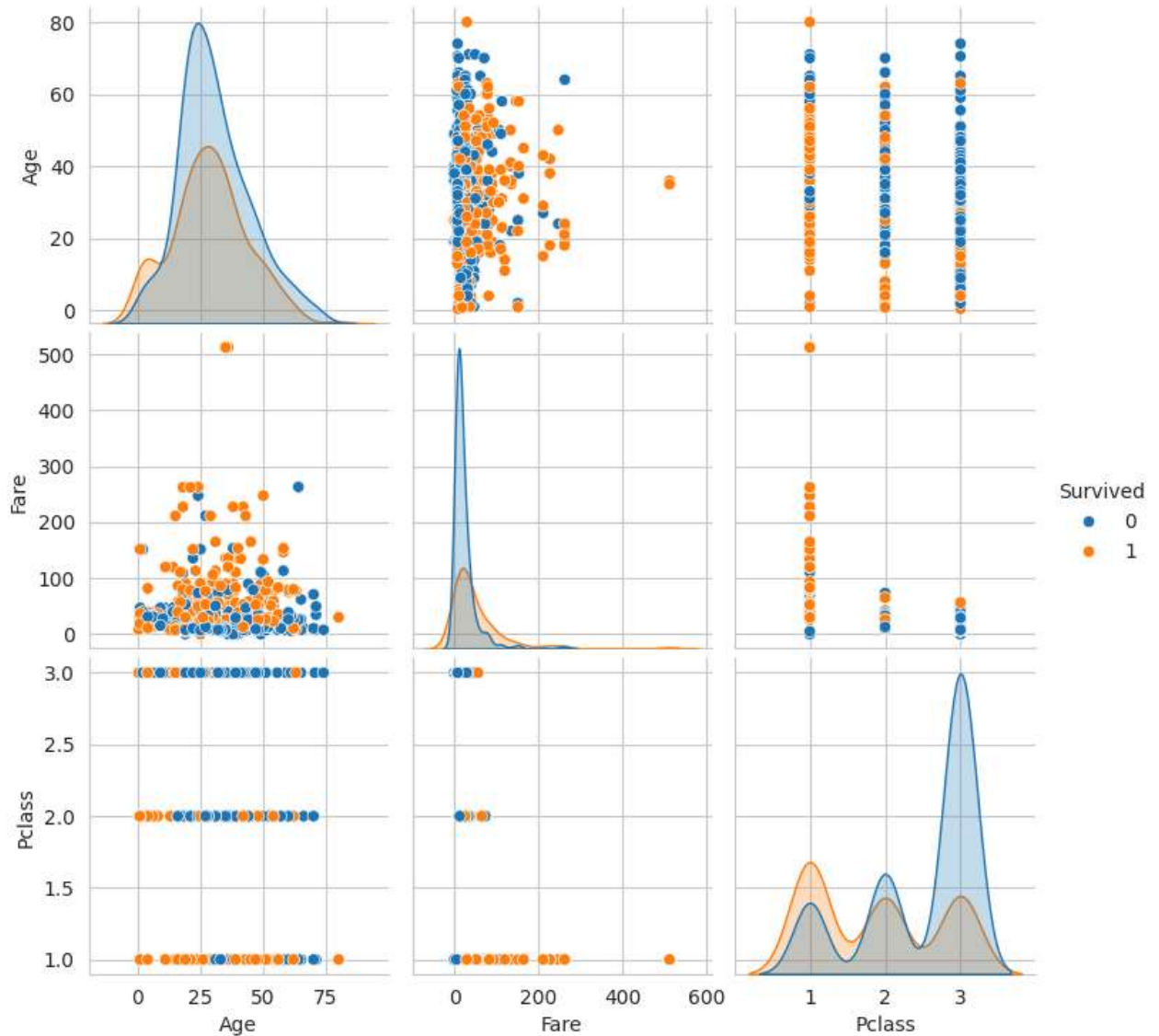
plt.title("Correlation Heatmap")
plt.show()
```



Observation

- Passenger class and fare show relationships with survival, while some variables have only weak correlations.

```
sns.pairplot(df[['Survived', 'Age', 'Fare', 'Pclass']].dropna(), hue='Survived')
plt.show()
```



Observation

- The pairplot highlights patterns between age, fare, passenger class, and survival status.

```
plt.figure(figsize=(8,5))
sns.scatterplot(x='Age', y='Fare', hue='Survived', data=df)
plt.title("Age vs Fare")
plt.xlabel("Age")
plt.ylabel("Fare")
plt.show()
```

